

Guangyu Meng

Ph.D. Candidate, University of Notre Dame

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RESEARCH INTERESTS

My research integrates topological data analysis (TDA) with deep learning for medical image analysis, organized around three themes: (1) embedding persistent homology into self-supervised and unsupervised learning for structurally aware representations, (2) scalable topological computation including GPU-accelerated persistent homology and optimal transport-based comparison of topological summaries, and (3) neuro-symbolic reasoning via LLM-based agentic frameworks for automated topological analysis. I also develop curriculum learning methods for efficient LLM fine-tuning. My long-term goal is topology-driven computational phenomics: trustworthy, auditable AI that characterizes biological structure from cells to whole organisms.

EDUCATION

University of Notre Dame

Ph.D. in Computer Science (GPA: 3.92/4.0)

South Bend, IN

2021 – Present (Expected Fall 2026)

- Focus: Topological Deep Learning for Medical Imaging & Efficient AI
- Advisors: Prof. Danny Z. Chen (**IEEE Fellow, ACM Distinguished Scientist**) & Prof. Erin W. Chambers
- Dissertation: *Efficient Analysis of Complex Data Using Machine Learning and Topological Methods: Theory, Computation, and Applications*

University of Notre Dame

M.S. in Computer Science

South Bend, IN

August 2025

Washington University in St. Louis

Graduate Studies in Computer Science (GPA: 4.0/4.0)

St. Louis, MO

2020 – 2021

- Transferred to the Ph.D. program at the University of Notre Dame.

University of Southampton

M.S. in Microelectronics System Design (With Merit)

Southampton, UK

2012

University of Birmingham

Bachelor of Engineering with Honours Class II (Division I) (Electronic Engineering)

Birmingham, UK

2010 – 2011

Huazhong University of Science and Technology

B.E. in Electronic Science and Engineering

Wuhan, China

2007 – 2010

RESEARCH EXPERIENCE

Graduate Research Assistant

Department of Computer Science and Engineering

University of Notre Dame

2021 – Present

- **TopoCL – Topological Contrastive Learning**: Designed a universal framework featuring topology-aware augmentations, a Hierarchical Transformer-based Topology Encoder, and an adaptive mixture-of-experts (MoE) fusion module for integrating visual and topological representations. Achieved +3.26% average accuracy improvement with strong statistical significance (80% of comparisons with $p < 0.001$) across five contrastive methods on five medical imaging datasets.
- **VTRL-Med – Medical Image Deep Clustering**: Developed a visual-topology framework for unsupervised medical image clustering, integrating SAM with mask-based structural priors and persistent homology features for anatomically consistent representations. Achieved state-of-the-art 81.51% accuracy across 13 benchmarks (+4.48% over baselines), nearly matching supervised methods.
- **NNS-EMD – Efficient Optimal Transport**: Proposed a GPU-accelerated Nearest Neighbor Search approximation for Earth Mover's Distance with comprehensive theoretical analysis of time complexity and error bounds. Achieved $44\times$ – $135\times$ speedup while improving accuracy (91.88% vs 87.42% on 20news, 93.24% vs 78.53% on Amazon reviews).
- **Reeb Graph Comparison – Computational Topology**: Proposed a provably stable framework for Reeb graph comparison using Gromov–Wasserstein distance with persistence images, establishing rigorous theoretical stability bounds under scalar field perturbations. Achieved superior classification accuracy (96.16% on ModelNet10, 100% on Mesh) over bottleneck distance and graph edit distance baselines.
- **CuPH – GPU-Accelerated Cubical Persistent Homology**: Developed algorithmic innovations (event-driven processing, hierarchical label propagation) and GPU engineering optimizations for cubical persistent homology. Achieved 14,444 images/sec on 224×224 inputs ($150\times$ faster than GUDHI, $361\times$ at 2048×2048) with $6.7\times$ lower GPU memory than the prior GPU baseline.

- **TopoAgent – Agentic Topology Learning:** Developed an LLM-based agentic framework that automates topology learning for medical image analysis using a Perception–Reasoning–Action–Reflection loop with 21 domain-specific tools, skill distillation from 15 topological descriptors across 26 datasets, and dual memory. Introduced TopoBenchmark (113,182 samples, 26 datasets, 5 object types). Achieved 68.21% average balanced accuracy, outperforming the strongest single-descriptor baseline by 9.32% and general-purpose LLMs by over 21%.
- **PUDF – Curriculum Learning for LLMs:** Developed a psychology-based curriculum learning framework applying Item Response Theory to artificial crowds for model-independent difficulty quantification and adaptive training scheduling. Achieved 4.13% accuracy improvement with 69.68% faster training on AG News (Llama3.1-8B), and 76.70% accuracy on GSM8K mathematical reasoning (Qwen2.5-7B) with 42% speedup.

PUBLICATIONS

Peer-Reviewed Journal Articles

- [J1] **Guangyu Meng**, Qingkai Zeng, John P. Lalor, Hong Yu, “Psychology-based Unified Dynamic Framework for Curriculum Learning,” *Computational Linguistics*, 2025. DOI: [10.1162/COLLa.584](https://doi.org/10.1162/COLLa.584)
- [J2] Peixian Liang, Yifan Ding, Yizhe Zhang, Jianxu Chen, Hao Zheng, Hongxiao Wang, Yeja Zhang, **Guangyu Meng**, Tim Weninger, Michael Niemier, X. Sharon Hu, Danny Z. Chen, “Cell Instance Segmentation: The Devil Is in the Boundaries,” *IEEE Transactions on Medical Imaging (TMI)*, 2025. DOI: [10.1109/TMI.2025.3621093](https://doi.org/10.1109/TMI.2025.3621093)
- [J3] **Guangyu Meng**, Ruyu Zhou, Liu Liu, Peixian Liang, Fang Liu, Danny Z. Chen, Michael Niemier, X. Sharon Hu, “Efficient Approximation of Earth Mover’s Distance Based on Nearest Neighbor Search,” *IEEE Transactions on Multimedia (TMM)*, 2025. DOI: [10.1109/TMM.2025.3623529](https://doi.org/10.1109/TMM.2025.3623529)

Peer-Reviewed Conference Papers

- [C1] **Guangyu Meng**, Pengfei Gu, Peixian Liang, John P. Lalor, Erin W. Chambers, Danny Z. Chen, “TopoCL: Topological Contrastive Learning for Medical Imaging,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [C2] **Guangyu Meng**, Qisheng Jiang, Kaiqun Fu, Beiyu Lin, Chang-Tien Lu, Zhiqian Chen, “Early Forecasting of the Impact of Traffic Accidents Using a Single Shot Observation,” *SIAM International Conference on Data Mining (SDM)*, 2022.

Manuscripts Under Review

- [R1] **Guangyu Meng**, Pengfei Gu, John P. Lalor, Peixian Liang, Erin W. Chambers, Danny Z. Chen, “VTRL-Med: Vision-Topology Reinforced Learning for Medical Image Clustering,” under review at *European Conference on Computer Vision (ECCV)*, 2026.
- [R2] **Guangyu Meng**, Pengfei Gu, Xueyang Li, Yiyu Shi, Erin W. Chambers, Danny Z. Chen, “TopoAgent: An Agentic Framework for Automated Topology Learning in Medical Imaging,” under review at *European Conference on Computer Vision (ECCV)*, 2026.
- [R3] **Guangyu Meng**, Erin W. Chambers, “Provably Stable Reeb Graph Comparison via Gromov–Wasserstein Distance,” under review at *Journal of Applied and Computational Topology*, 2026.

PRESENTATIONS

- “Provably Stable Reeb Graph Comparison via Gromov–Wasserstein Distance,” Conference on Topological Data Analysis (TDA 2025), University of Missouri-Columbia, November 2025. (Oral talk and poster)

TEACHING EXPERIENCE

Teaching Assistant

Department of Computer Science and Engineering

University of Notre Dame

2021 – Present

- CSE 60111: Complexity and Algorithms, Spring 2025
- CSE 20110: Discrete Mathematics, Fall 2024 – Designed two course assignments
- CSE 40625/60625: Machine Learning, Fall 2021 – Delivered a tutorial on PyTorch
- CSE 30321: Computer Architecture, Spring 2021

MENTORING

Xianshi Ma (xma3@nd.edu), Ph.D. student, University of Notre Dame (advised by Prof. Danny Z. Chen) – Model fusion for semi-supervised medical video segmentation

Colby Nelson (cnelso23@nd.edu), Master’s student, University of Notre Dame – Deep clustering for medical imaging

INDUSTRY EXPERIENCE

Emulation Specialist

Shenzhen, China

Mentor Graphics Co., Ltd

2016 – 2019

- Led high-performance verification solutions for Huawei's first commercial 5G chipset; optimized testing code, reducing runtime from 45 to 8 minutes through parallelization.
- Delivered technical presentations at Silicon Valley headquarters and led training for FiberHome on UVM environment configuration.

ASIC Design Engineer

Shanghai, China

Spreadtrum Communications

2015 – 2016

- Maintained chip-level emulation environments; developed Python-based resource allocation program improving device utilization by 20%.

NPI Engineer

Shanghai, China

NXP Semiconductors

2013 – 2015

- Optimized wafer mapping and packaging processes, enhancing die bond machine capacity by 14.8%.

TECHNICAL SKILLS

Core Domains: Artificial Intelligence, Medical Image Analysis, Semiconductor Verification (EDA)

Deep Learning: PyTorch, TensorFlow, LLM Fine-tuning (LoRA/QLoRA), Transformers, SAM, Vision Transformers

LLM & Agentic AI: Hugging Face, LangChain, LangGraph, Prompt Engineering, Tool-Augmented LLM Agents

Mathematics: Topological Data Analysis (TDA), Persistent Homology, Optimal Transport

Programming: Python, C/C++ , CUDA (High-Performance GPU Computing), MATLAB, SystemVerilog

Systems & Tools: Linux/HPC (Slurm), Docker, Git, Mixed-Precision Training, Multi-GPU Distributed Training

PROFESSIONAL SERVICE

Conference Reviewer: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026, Symposium on Computational Geometry (SoCG) 2025, AAAI Conference on Artificial Intelligence 2024, SIAM International Conference on Data Mining (SDM) 2023

Journal Reviewer: Computing in Geometry and Topology (CGT)

REFERENCES

Erin W. Chambers (Co-advisor)

Snyder Family Mission Collegiate Professor

Department of Computer Science and Engineering

University of Notre Dame

echambe2@nd.edu

Danny Z. Chen (Co-advisor)

Professor, IEEE Fellow, ACM Distinguished Scientist

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X. Sharon Hu (Collaborator)

Leo E. and Patti Ruth Linbeck Professor

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University of Notre Dame

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Pengfei Gu (Collaborator)

Assistant Professor

Department of Computer Science

University of Texas Rio Grande Valley

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John P. Lalor (Collaborator)

Assistant Professor, Mendoza College of Business

Concurrent Professor, Computer Science and Engineering

University of Notre Dame

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